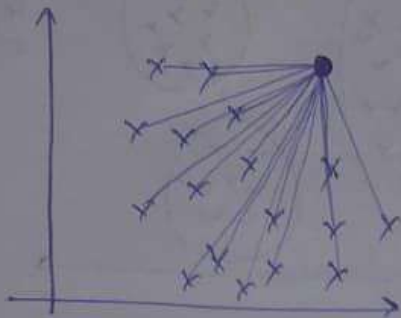
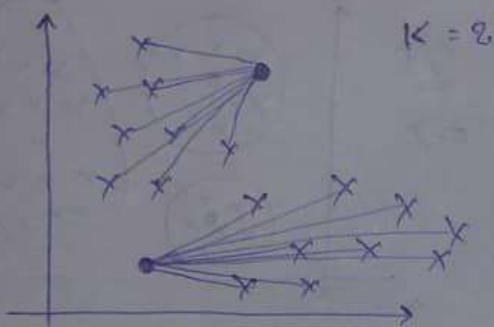


wcss = within cluster sum of squares

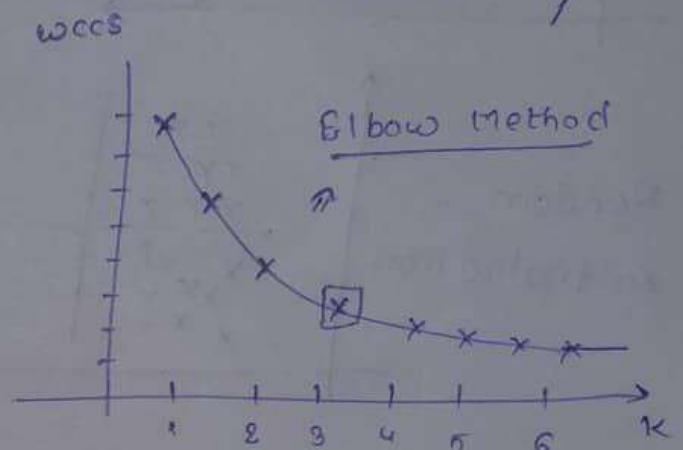
Initialize $K = 1$ to 20



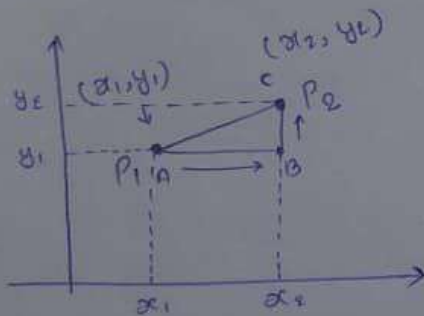
$$wcss = \sum_{i=1}^n (\text{distance between point to nearest centroid})^2$$



wcss ↓ ↓



* Eucledian Distance



$$\text{Euclidean dist.} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Manhattan dist.} = |x_2 - x_1| + |y_2 - y_1|$$

IRON MAN → U.S



⇒ Manhattan distance

Air Traffic

